

Computer Science & IT

Database Management System



Query Languages

Lecture No. 01



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Recap of Previous Lecture

- ✓ **Topic** Properties of multivalued dependency
- ✓ **Topic** Fourth normal form (4NF)



Topics to be Covered



Topic

Query Languages



Topic

Relational algebra



Topic

Basic relational algebra operations





Topic : Query languages

Query languages

Procedural query language

We define, what needs to be retrieved from the database, along with the procedure to retrieve the data from database

eg: Relational Algebra

Non-procedural query language

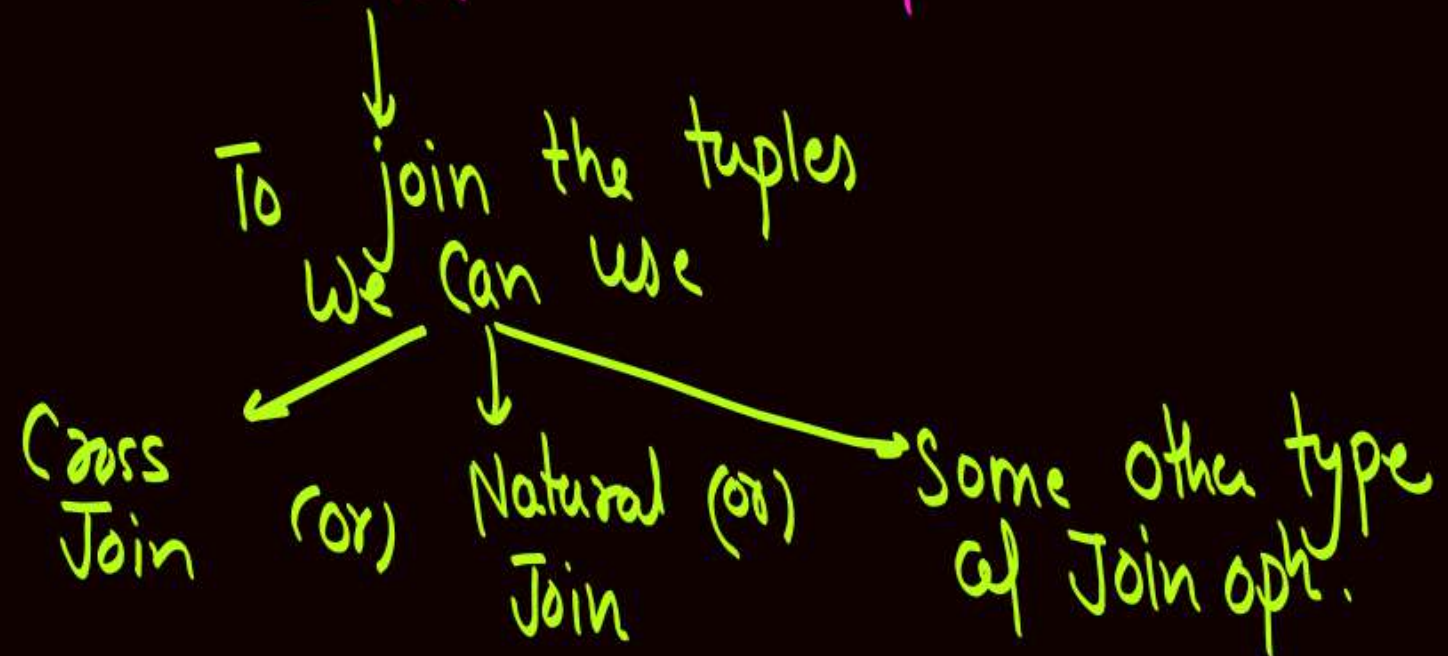
We only define, what needs to be retrieved from the database, and we use the syntax provided by that query language to retrieve the information from the database

* eg: SQL
Tuple Relational Calculus (TRC)
Domain Relational Calculus (DRC)

Note:-

Query condition evaluates tuple by tuple, only one tuple at a time.

→ ∴ If we want to compare two or more tuples of the same table or different tables, then we need to join those tuples into a single tuple



Note:- Relational algebra query will always produce distinct tuples.



Topic : Relational Algebra

- ✓ Relational Algebra is a procedural query language used to query the relational database tables to access data.

- * Relational Algebra operation can be classified into two types:
 - ✓ 1) Basic Relational Algebra Operations
 - ✓ 2) Derived Relational Algebra operations



Topic : Basic Relational Algebra operators

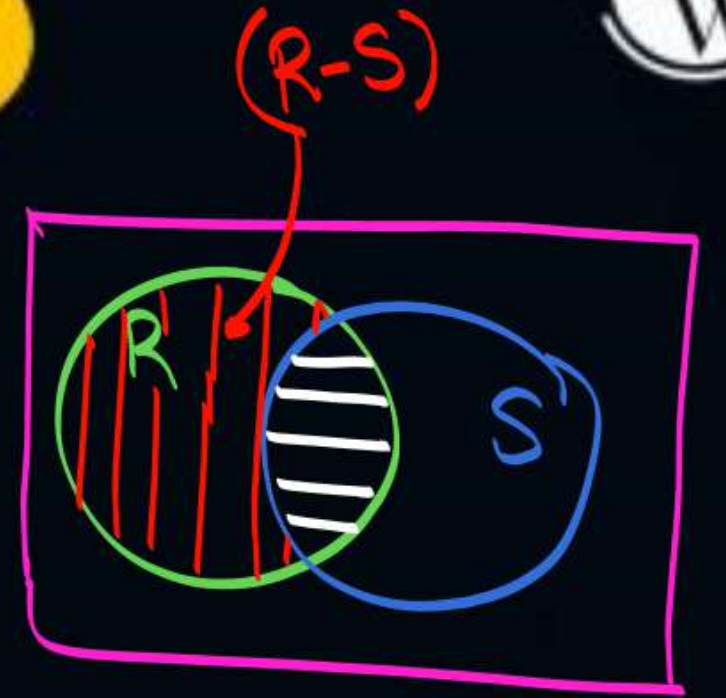
- ✓ 1. Projection(π) ✓
 - ✓ 2. Selection (σ) ✓
 - ✓ 3. Cross Product ($\times$) ✓
 4. Union (U) ✓
 5. Set Difference ($-$) ✓
 6. Rename (ρ) ✓
- We have already discussed
when we were discussing
"natural join" oph.
- Set operations



Topic : Derived Relational Algebra operators

- ✓ 1. Intersection ($\cap$)
- ★ ✓ 2. Join Operations (" $\bowtie$ ")
- ★ ✓ 3. Division Operation ($\div$)

$$(R \cap S) = R - (R - S)$$





Topic : Projection (π)

- It is used to project the column data from a relation based on the attributes specified with projection operation.

e.g., $\pi_{\langle \text{attribute list} \rangle}(R)$

Attributes/Columns required in the O/p

Name of relation

Projection operator need not obey commutative property
i.e.

$$\pi_{\langle \text{list2} \rangle}(\pi_{\langle \text{list1} \rangle}(R)) \neq \pi_{\langle \text{list1} \rangle}(\pi_{\langle \text{list2} \rangle}(R))$$

(Sid) (Sid, Cid) Sid, Cid Sid

R

Sid	Cid	Branch
S ₁	C ₁	CS
S ₂	C ₂	IT
S ₃	C ₁	CS

①

$$\pi_{\text{Sid, Cid, Branch}}(R) \equiv (R)$$

o/p will be exactly same as relation R.

If projection opⁿ is not present, then all attributes/columns of the relation will be present in o/p

② Retrieve Sids of all the Students from relation R.

$$\pi_{\text{Sid}}(R) \Rightarrow \text{o/p} =$$

Sid
S ₁
S ₂
S ₃

③ Retrieve all Cids from the relation R.

$$\pi_{\text{Cid}}(R) \Rightarrow \text{o/p} =$$

Cid
C ₁
C ₂
C ₁

← duplicate tuples are not allowed

R

Sid	Cid	Branch
S_1	C_1	CS
S_2	C_2	IT
S_3	C_1	CS

④

$$\pi_{\text{Sid}, \text{Cid}}(R) \Rightarrow \text{o/p} =$$

Sid	Cid
S_1	C_1
S_2	C_2
S_3	C_1

tuple wise
these are
distinct
tuples.

$$\underbrace{\pi_{\text{Sid}}(\pi_{\text{Sid}, \text{Cid}}(R))}_{\text{it will execute}} \neq \underbrace{\pi_{\text{Sid}, \text{Cid}}(\pi_{\text{Sid}}(R))}_{\text{invalid R.A. query.}}$$



Topic : Selection(σ)

It is used to select the tuples from underlying relation based on the predicate condition specified with selection operation.

$$\sigma_{\langle \text{selection_condition} \rangle}(R)$$

Predicate Condⁿ

Name of relation

NOTE:

$$\sigma_{A \wedge B}(R) = \sigma_{B \wedge A}(R)$$

OR

$$\sigma_B(\sigma_A(R)) = \sigma_A(\sigma_B(R))$$

R

Sid	Cid	Branch
S ₁	C ₁	CS
S ₂	C ₂	IT
S ₃	C ₁	CS

①

$\pi_{\text{Sid, Cid, Branch}}(R)$

We did not specify any selection Condⁿ.
 ∴ All the tuples of the relation
 will be present in o/p

②

Select the records corresponding to Cid = "C₁"

$\sigma_{\text{Cid} = \text{C}_1}(R) =$

Sid	Cid	Branch
S ₁	C ₁	CS
S ₃	C ₁	CS

No projection
 opⁿ is present.
 ∴ All the attributes will be
 present in o/p

only the tuples in which
 Cid is "C₁" will
 be present in o/p.

R

Sid	Cid	Branch
S ₁	C ₁	CS
S ₂	C ₂	IT
S ₃	C ₁	CS

$\sigma_{Cid=C_1}$

(R) $\Rightarrow$ o/p =

Sid	Cid	Branch
S ₁	C ₁	CS
S ₃	C ₁	CS

③ Select Sids of the students who enrolled for course with Cid = C₁

$\pi_{Sid}(\sigma_{Cid=C_1}(R)) \Rightarrow$

Sid
S ₁
S ₃

R

Sid	Cid	Branch
S ₁	C ₁	CS ×
S ₂	C ₂ ×	IT
S ₃	C ₁	CS ×

④ Retrieve records
in which Cid=C₁
and Branch='IT'

Sid	Cid	Branch

$\sigma_{(Cid=C_1 \wedge Branch='IT')}(R) \Rightarrow \text{o/p will be Empty relation}$
 $\text{Cond}^n \rightarrow \sigma_{Branch='IT'}(\sigma_{Cid=C_1}(R))$

④ Retrieve records
in which Cid=C₁
or Branch='IT'

$\sigma_{(Cid=C_1 \vee Branch='IT')}(R) \Rightarrow \text{o/p} =$
 $\text{Cond}^n \rightarrow$

Sid	Cid	Branch
S ₁	C ₁	CS
S ₂	C ₂	IT
S ₃	C ₁	CS



Topic : NOTE



$$\rightarrow \sigma_{A \wedge B}(R) = \sigma_{B \wedge A}(R)$$

OR

$$\sigma_B(\sigma_A(R)) = \sigma_A(\sigma_B(R))$$

OR

$$\sigma_A(R) \cap \sigma_B(R)$$

(OR)

$$\sigma_B(R) \cap \sigma_A(R)$$

$$\sigma_{A \vee B}(R) \equiv \sigma_{B \vee A}(R)$$

(OR)

$$\sigma_A(R) \cup \sigma_B(R)$$

(OR)

$$\sigma_B(R) \cup \sigma_A(R)$$



Topic : Cross Product ($\times$)

/ Cross Join / Cartesian Product .



Cross-product is a binary operation. Let R and S are any two relation, then cross product $R \times S$ will result in all attributes of R followed by all attribute of S with all possible combinations of tuples from R and S.

x attributes

m tuples

R		
Sid	Cid	Branch
S ₁	C ₁	CS
S ₂	C ₂	IT
S ₃	C ₁	CS

y attributes

n tuples

S	
Sid	Sname
S ₁	Ram
S ₂	Mohan
S ₃	Ram

$R \times S =$

$(m \times n)$ tuples

$x+y$ attributes

R.Sid	R.Cid	R.Branch	S.Sid	S.Sname
S ₁	C ₁	CS	S ₁	Ram
S ₁	C ₁	CS	S ₂	Mohan
S ₁	C ₁	CS	S ₃	Ram
S ₂	C ₂	IT	S ₁	Ram
S ₂	C ₂	IT	S ₂	Mohan
S ₂	C ₂	IT	S ₃	Ram
S ₃	C ₁	CS	S ₁	Ram
S ₃	C ₁	CS	S ₂	Mohan
S ₃	C ₁	CS	S ₃	Ram



Topic : Union, Set difference, Intersection

- ❑ Union, Set Difference and Intersection are the Set operations.
- ❑ To use set theory operators on any two relations, those relations must be union compatible.
- ❑ The union compatibility of relations implies that the participating relations must fulfil the following conditions.
 1. Same degree, i.e. The two relations must have the same number of attributes.
 2. Same domain of each corresponding attributes of relations
"data type"

R

Sid	Cid	Branch

No. of attributes in Relation R $\neq$ No. of attributes in relation S.

S

Sid	Sname

∴ Given relation R & S are not Union compatible.

R

Sid	marks

int int

S

Sid	Sname

int Char

① No. of attributes in Relation R = No. of attributes in relation S.

i.e., first condition for union Compatibility is satisfied.

② Domain of first attribute of relation R is same as domain of first attribute of relation S

Because of this R & S are not Union Compatible

{ Both are integer type }
but domain of 2nd attribute of relⁿ R is not same as domain of 2nd attribute of relⁿ S.

R

Sid	Branch

S

Sid	Sname

Given Relation R & S
are Union Compatible.

① No. of attributes in Relation R = No. of attributes in Relation S.

i.e., first Condition for Union Compatibility is satisfied.

② Domain of first attribute of Relation R is same as domain of first attribute of Relation S
{ Both are integer type }

And domain of 2nd attribute of Relⁿ R is also same as domain of 2nd attribute of Relⁿ S.

2nd Condition is also satisfied

Note: ① If relations are Union Compatible, then set operations can be performed on those relation.

And

② After the set operation, the resulting relation will take the names of its attributes from left hand side relation.

ie in " $R \cup S$ ", names of attributes will be same as names of attributes in relation R .

And in " $S \cup R$ "; names of attributes will be same as names of attributes in relation " S ".



Topic : Union, Set difference, Intersection

- ❑ **Union ($A \cup B$)**- It contains unique tuples from both the relations.
- ❑ **Difference ($A - B$)**- It contains all the tuples that are contained in the relation A but are not present in the relation B
- ❑ **Intersection ($A \cap B$)**- It contains all the tuples that are contained in both the relations A as well as in relation B.

Employee (E)

Eid	Ename
1	A
3	B
5	A
8	C

EUS =

Eid	Ename
1	A
3	B
5	A
8	C
2	D
7	C

SNE =

|||
S-(S-E)

Sid	Sname
3	B
5	A

Student (S)

Sid	Sname
2	D
3	B
5	A
7	C

S-E =

Sid	Sname
2	D
7	C

E-S =

Eid	Ename
1	A
8	C



Topic : Rename (ρ)

Student(Student-id, Student-name)



Rename operation can be used to rename attribute of the relation, name of the relation or both.

* Renaming a relation:

$\rho_{\text{FinalYrStudents}}(\text{Students})$

Rename opn

Current name of relation

Names of attributes will remain same

New name assigned to relation

* Renaming attributes:

$\rho_{(\text{SID}, \text{Sname})}(\text{Students})$

Current name of relation

Name of relation will remain same

New name for first attribute

New name for second attribute

* Renaming both:

$\rho_{\text{FinalYrStudents}}(\text{SID}, \text{Sname})(\text{Students})$

New Name for relation

Current name of relation



2 mins Summary



Topic

Query Languages ✓

Topic

Relational algebra ✓

Topic

Basic relational algebra operations ✓

THANK - YOU